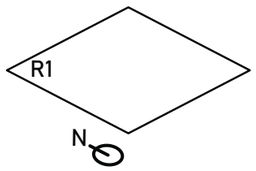
Computational Design Workflow

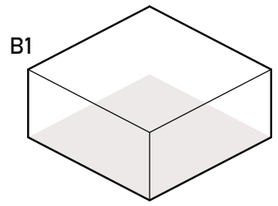
Define base rectangle R1 in the XY plane

R1 : X = 13, Y = 13



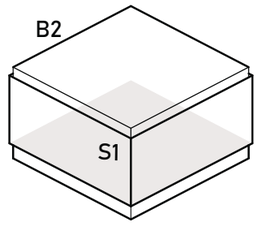
Extude R1 using 'Box Rectangle' in Z direction

B1 : Rectangle = R1, Height = 7



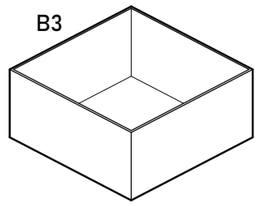
'Deconstruct Brep' B1, 'Construct plane' for S1. Create B2 using 'Centre Box'

B2 : X = Y = 6.3 , Z = 15.945



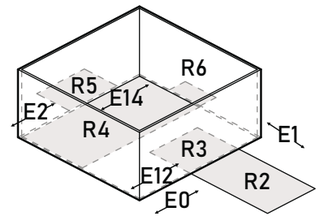
Create B3, using 'Solid Difference' subtracting B2 from B1

B3 : Brep A = B1, Brep B = B2



Define base rectangles R2, R3, R4, R5, R6 in the XY plane relative to B3 edges

R2 : X = -8.2, Y = -5.1, E0 = 10.6 R3 : X = -5.1, Y = 4.2, E12 = 10.6
R4 : X = 12.6, Y = -6.4, E14 = -0.2 R5 : X = -5.1, Y = 3.5, E2 = 2.4
R6 : X = 1.7, Y = 3.3, E1 = 6.5



Extude R2, R3, R4, R5, R6 using 'Box Rectangle' in Z direction

B4 : Rectangle = R2, Height = 6.5 B5 : Rectangle = R3, Height = 6
B6 : Rectangle = R4, Height = 6 B7 : Rectangle = R5, Height = 6.5
B8 : Rectangle = R6, Height = 2.7

Create a reference geometry using 'Polyline' connecting 3 points, moving P2 to create P3

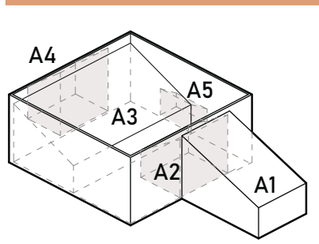
P3: Geometry = P2, Motion = -3.33 (Unit Z)
L1 : Vertices = P1 P2 P3, Closed = True

Extrude L1 in X direction, distance is width of R2

C1: Base = L1, Direction = 5.1 (Unit X)

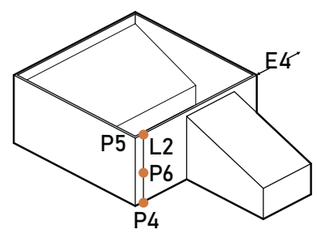
Repeat previous 2 steps for B5, R6, R7, R8

Create angled geometry, using 'Solid Difference' subtracting Cs from B(4-8)



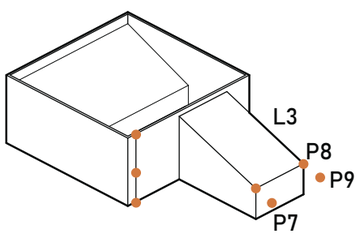
Create roof connection geometry on B3, at point P6 on a polyline connecting two edges

P4: Curve = E0, Parameter = 0.615
P5: Curve = E4, Parameter = 0.615
L2: Vertices = P4 P5, Closed = True
P6: Curve = L2, Parameter = 0.583



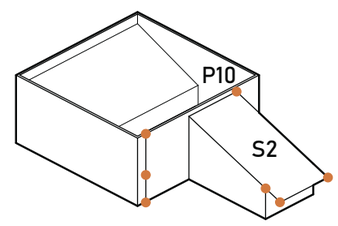
Make external roof geometry reference point, using 'Move' for P3 along L1 and P8 along L3

P7: Geometry = P3, Motion = L1 (Amplitude = 1.7)
P9: Geometry = P8, Motion = L3 (Amplitude = 1.7)



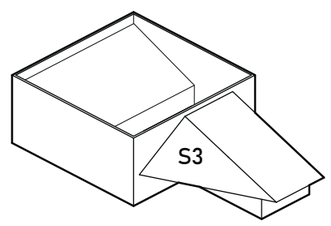
Using '4 Point Surface' generate S2

S2: Corner A = P1, Corner B = P10, C = P9, Corner D = P7

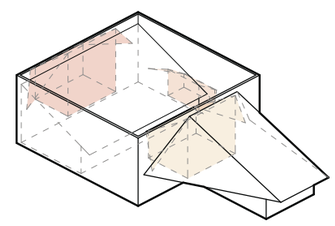


Using '4 Point Surface' generate S3

S3: Corner A = P1, Corner B = P7, C = P6



Repeat previous 4 steps for all roof surfaces



The key design concepts when massing the DHY house are the main box geometry encasing the internal courtyard and the unique shaped roofs that promote shading. The inputs to generate the main box geometry (B3) are fixed constraints keeping the prominent volume the same for all future variations. Contrastingly, the roof geometry has remained variable both in angle and size. This allows variations of the roof geometry to be explored, manipulating the Vietnamese sun while retaining the precedent's values of shared communal space (Abdel, 2023).

In terms of industry use, completing this massing stage parametrically generates a flexible workflow built upon rapid variations with instant visual feedback. Not only is this time and cost effective but also produces high quality end results.

Legend

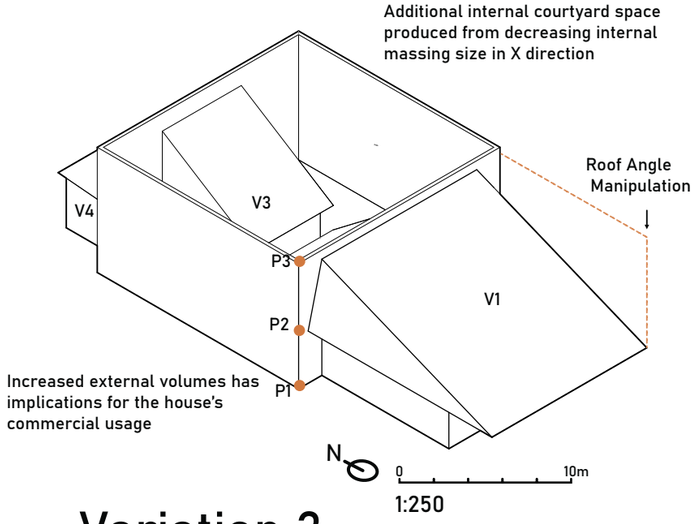
Prefixes:
B = Box
R = Rectangle
E = Edge
P = Point
L = Line
C = Cutting Geometry
A = Angled Shape
S = Surface

Key Steps
Variables
Fixed Constraints

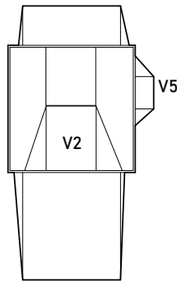
Massing Model Variations

Variation 2

- Switch internal and external massing dimensions
- Extend south external roof geometry to account for additional shade requirements



Top View

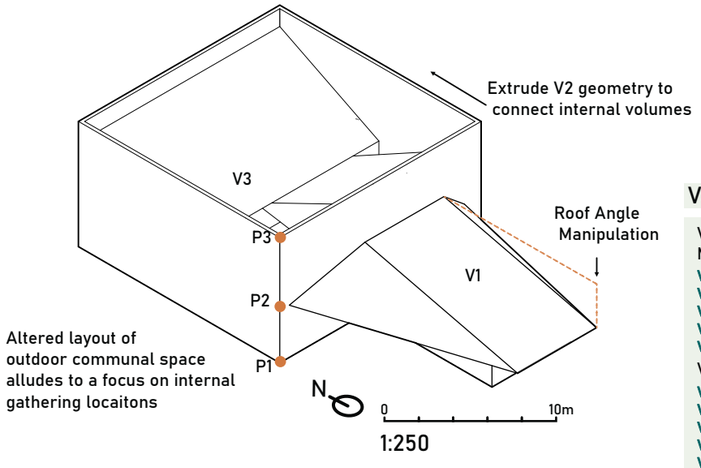


Variable Parameter Inputs

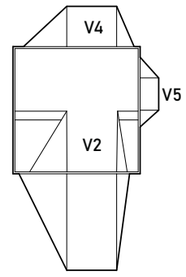
Volume Position on Main Box Geometry	Roof Angle Manipulation
V1: 11.5m	V1: -3.33 | V4: -3.2
V2: 9m	V2: -3.2 | V5: -1.3
V3: -4m	V3: -3.2
V4: 11.5m	
V5: 6.5m	Roof Surface Location
Volume Size	V1: East = [P1 = 11.9 P2 = 0.58 P3 = 11.9]
V1: [X = 10 Y = 8.2]	West = [P1 = 0.615 P2 = 0.58 P3 = 0.615]
V2: [X = 5.1 Y = 4.2]	V2: East = [P1 = 11.68 P2 = 0.37 P3 = 11.68]
V3: [X = 5.1 Y = 5.1]	West = [P1 = 1.705 P2 = 0.37 P3 = 1.705]
V4: [X = 10 Y = 3.5]	V3: Centre = [P1 = 0 P2 = 1 P3 = 0]
V5: [X = 1.7 Y = 3.5]	V4: East = [P1 = 0.615 P2 = 0.58 P3 = 0.615]
	West = [P1 = 11.9 P2 = 0.58 P3 = 11.9]
	V5: North = [P1 = 11.9 P2 = 0.75 P3 = 11.9]
	South = [P1 = 4.795 P2 = 0.75 P3 = 4.795]

Variation 3

- Connect internal volumes
- Shrink external volumes to be one story and flare the roof



Top View

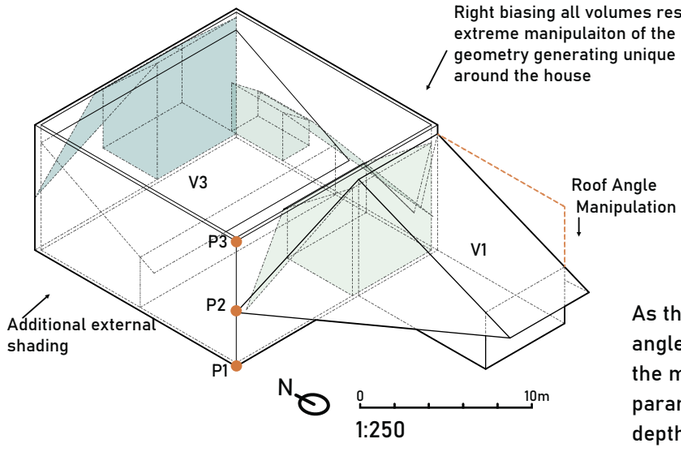


Variable Parameter Inputs

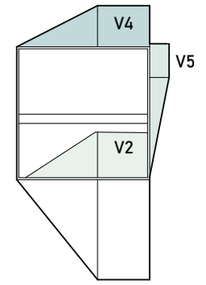
Volume Position on Main Box Geometry	Roof Angle Manipulation
V1: 10.6m	V1: -2 | V4: -2
V2: 10.6m	V2: -3.2 | V5: -1.3
V3: 0m,	V3: -3.2
V4: 10.6m	
V5: 6.5m	Roof Surface Location
Volume Size	V1: East = [P1 = 11.9 P2 = 0.58 P3 = 11.9]
V1: [X = 5.1 Y = 8.2]	West = [P1 = 0.615 P2 = 0.58 P3 = 0.615]
V2: [X = 5.1 Y = 6.6]	V2: East = [P1 = 11.68 P2 = 0.37 P3 = 11.68]
V3: [X = 12.6 Y = 6.4]	West = [P1 = 1.705 P2 = 0.37 P3 = 1.705]
V4: [X = 3.5 Y = 5.1]	V3: Centre = [P1 = 0 P2 = 3 P3 = 0]
V5: [X = 1.7 Y = 3.3]	V4: East = [P1 = 0.615 P2 = 0.58 P3 = 0.615]
	West = [P1 = 11.9 P2 = 0.58 P3 = 11.9]
	V5: North = [P1 = 11.9 P2 = 0.75 P3 = 11.9]
	South = [P1 = 4.795 P2 = 0.75 P3 = 4.795]

Variation 1

- Right bias volumes
- Extend all left roof geometry and connect to relative bottom edge of central box, shading from various sun angles



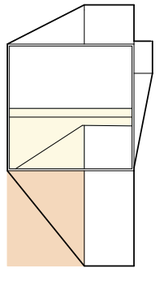
Top View

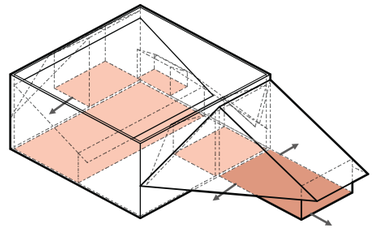


Variable Parameter Inputs

Volume Position on Main Box Geometry	Roof Angle Manipulation
V1: 13m	V1: -3.33 | V4: -3.2
V2: 12.8m	V2: -3.2 | V5: -1.3
V3: 0m	V3: -3.2
V4: 0m	
V5: 10m	Roof Surface Location
Volume Size	V1: East = [P1 = 0 P2 = 0 P3 = 0]
V1: [X = 8.2 Y = 5.1]	West = [P1 = 0 P2 = 0.5 P3 = 0]
V2: [X = 5.1 Y = 4.2]	V2: East = [P1 = 0 P2 = 0 P3 = 0]
V3: [X = 12.6 Y = 6.4]	West = [P1 = 0 P2 = 0.3 P3 = 1]
V4: [X = 5.1 Y = 3.5]	V3: Centre = [P1 = 0 P2 = 0.5 P3 = 0]
V5: [X = 1.7 Y = 3.3]	V4: East = [P1 = 0 P2 = 0 P3 = 0]
	West = [P1 = 13 P2 = 0.5 P3 = 13]
	V5: North = [P1 = 0 P2 = 0 P3 = 0]
	South = [P1 = 0 P2 = 0.75 P3 = 1.2]

Programatic Analysis





As the key design variables are the roof angles and the volumes connected to the main box geometry, altering them parametrically produces the most in-depth variations. The chosen variation's right-biasing retains the concept of communal shaded space but shift its location. This offers more external commercial shaded space.

Variation 2 further explores the volumetric impact via switching the internal and external volume size. This starkly changes the communal gathering spaces. Variation 3 acts to subtract the 'porch' gathering space by connecting the internal geometry, again shifting the space's core spatial roles. (Barandy, 2023)

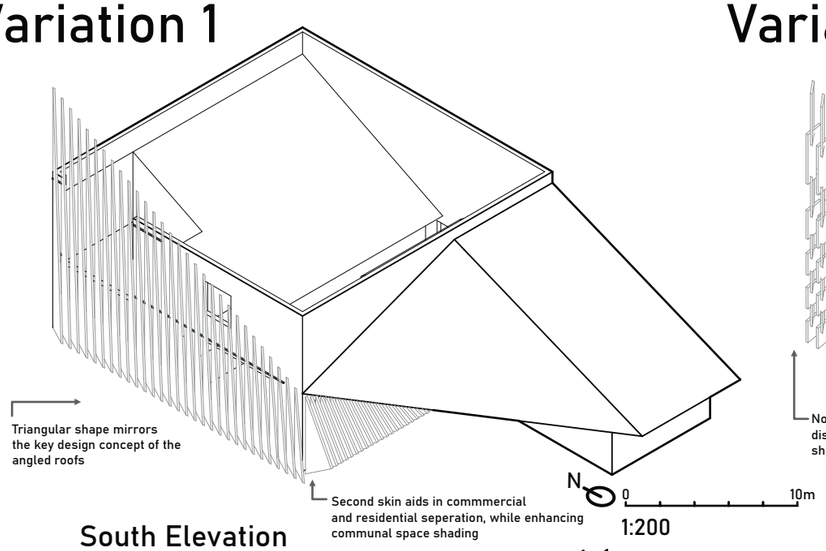
Legend

Prefixes:
V = Volume
P = Point

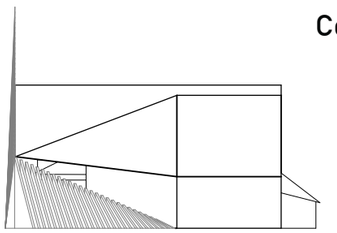
- Outdoor Communal
- Outdoor Commercial
- Indoor Communal
- Indoor Commercial
- → Shifted External Access

Second Skin Variations

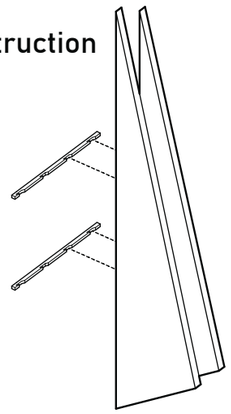
Variation 1



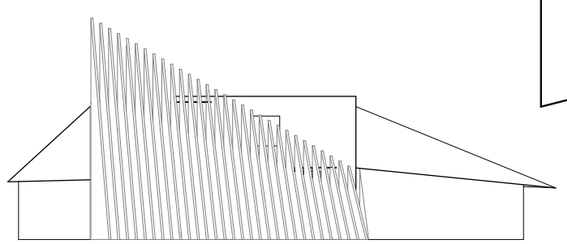
South Elevation



Construction



West Elevation



Parameter Inputs

Louvre Divisons

Uniform Sub Division: U = 30, V = 1.42

Louvre Extrusion

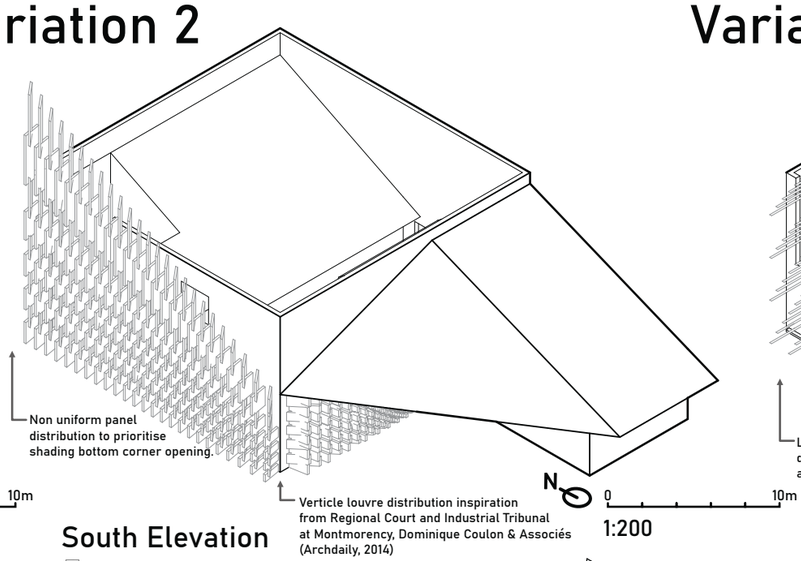
Extrusion: Triangle, Amplitude = 1

Extrusion Manipulation

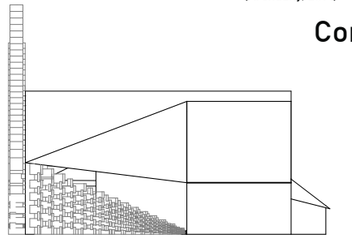
Rotate Plane: Angle = 64 (degrees)

The second skin louvres for variation 1 achieve a cohesive architectual design, mirroring the key roofing concepts of DHL House. The west facing wall in combination with the 64 degree angle louvre will greatly impact the shading into the encased courtyard depending on the time of day and time of year. Having this parameter as a varible simplified and optimised the design process for this variation, making it evident why parametric design is used in industry.

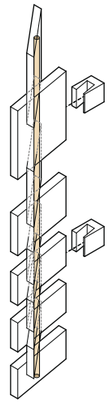
Variation 2



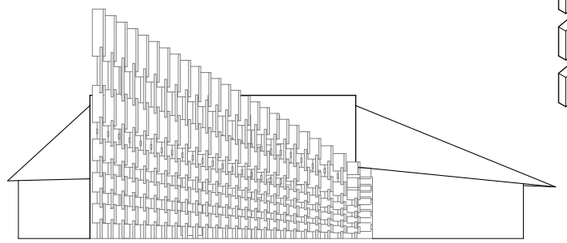
South Elevation



Construction



West Elevation

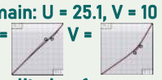


Parameter Inputs

Louvre Divisons

Non Uniform: Divide Domain: U = 25.1, V = 10

Domain: U= V =



Louvre Extrusion

Extrusion: Rectangle, Amplitude = 1

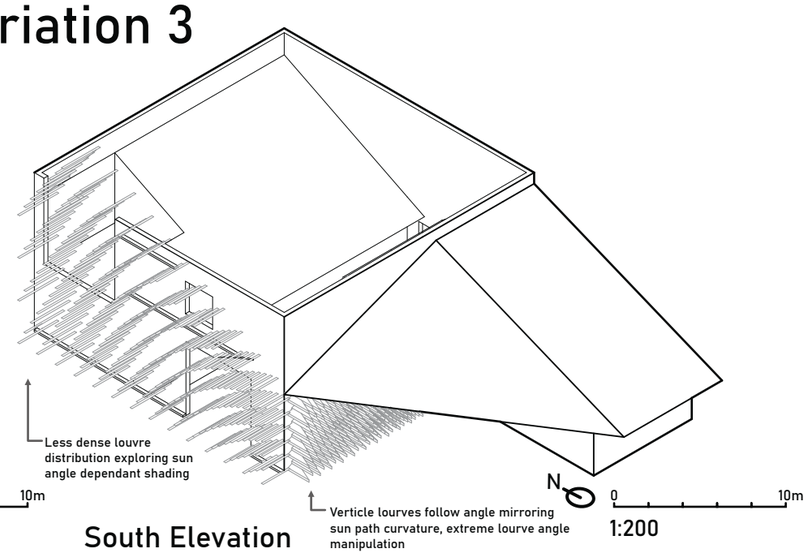
Extrusion Manipulation

Lists Rotation: Data = (10, 40, 10, 40, 10, 40, 10, 40)

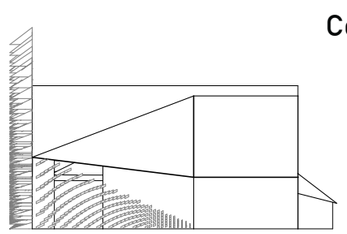
Scaling: Plane = (0;0), Scale X = 0.83, Scale Y = 1.25

Variation 2 utilises Graph-Mapper U and V inputs to attract louvres to the bottom right corner, shading the prominent opening. This is inspired by the lourve formating on Regional Court and Industrial Tribunal at Montmorency, Dominique Coulon & Associés. Where natural light is optimised with specifically angled vertical louvres.

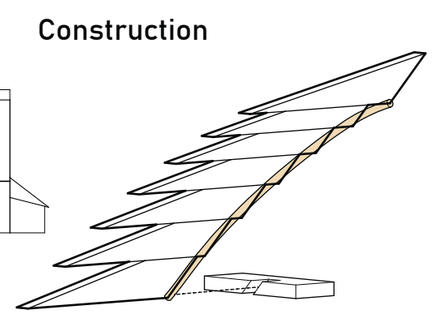
Variation 3



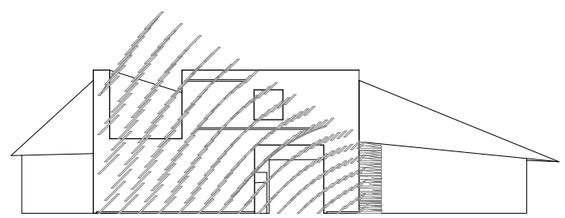
South Elevation



Construction



West Elevation



Parameter Inputs

Louvre Divisons

Dispatch: Divide Domain: U = 40, V = 17.40

Dispatch Pattern: True, False, False

Louvre Extrusion

Extrusion: Triangle, Amplitude = 1

Extrusion Manipulation

Scaling: Plane = (0;0), Scale X = 1, Scale Y = 2

Rotate Axis: Angle = -45

Variation 3 takes a more extreme approach in manipulating the shading into the DHY House's communal courtyard and commercial external courtyard. The Variation utilised dispatch when subdividing, allowing for the curvature of the vertical lourves. Introducing more shade from higher sun angles.

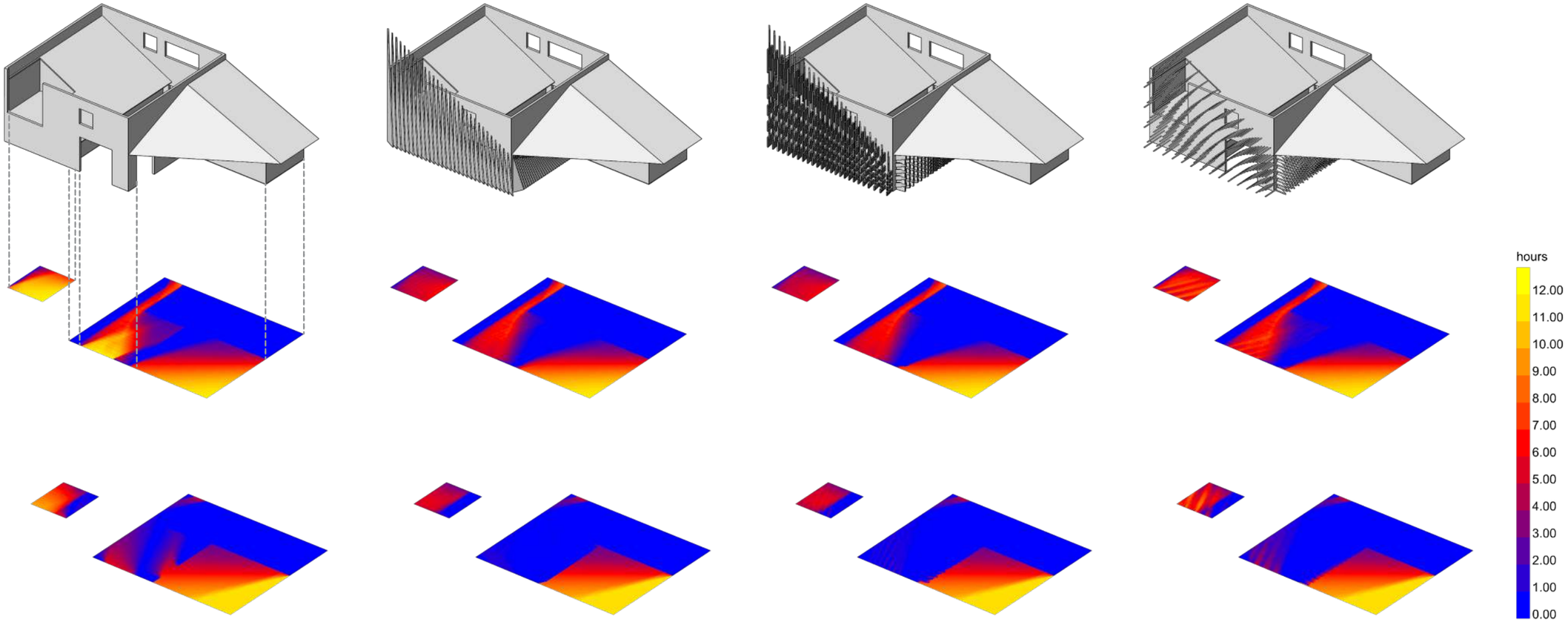
Solar Analysis

Control

Second Skin 1

Second Skin 2

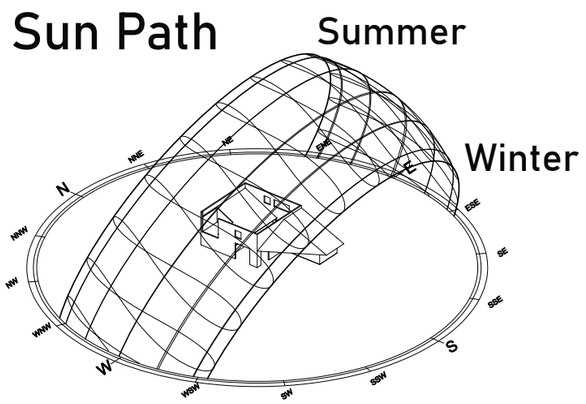
Second Skin 3



All variations balance effective shading performance in the precedent's environment with the visual design intent. Analysing the floor plates it can be seen that all variations produce similar results for the ground level sun hours results during winter solstice. However, Second Skin 3 achieves some breakthrough sun of approximately 3 hours which aligns with DHY House's key design concept of having natural lighting to promote communal spaces. Contrasting this optimal performance is however the variation's level 1 results in which the louvre angle is ineffective for the sun's path.

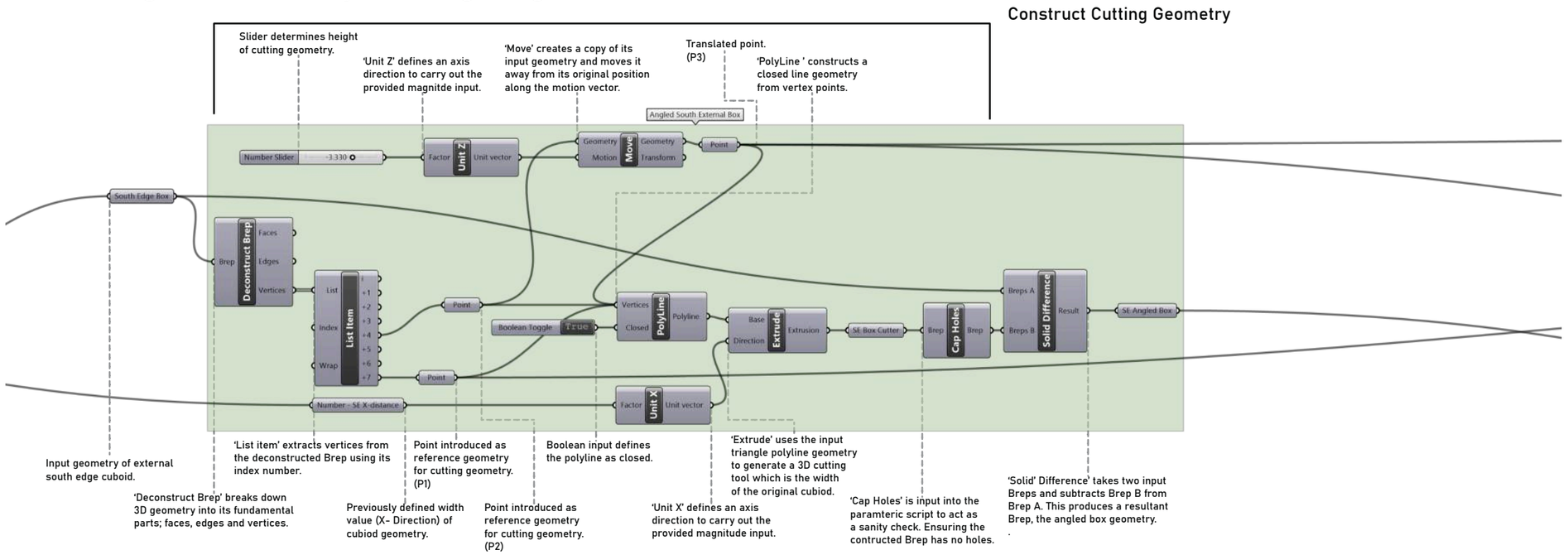
During summer solstice, Second Skin 1 achieves the best balance of internal courtyard shading. Allowing for natural light on the veranda communal space in combination with shading to promote natural cooling in Vietnam's hot environment. (Abdel, 2023).

Sun Path



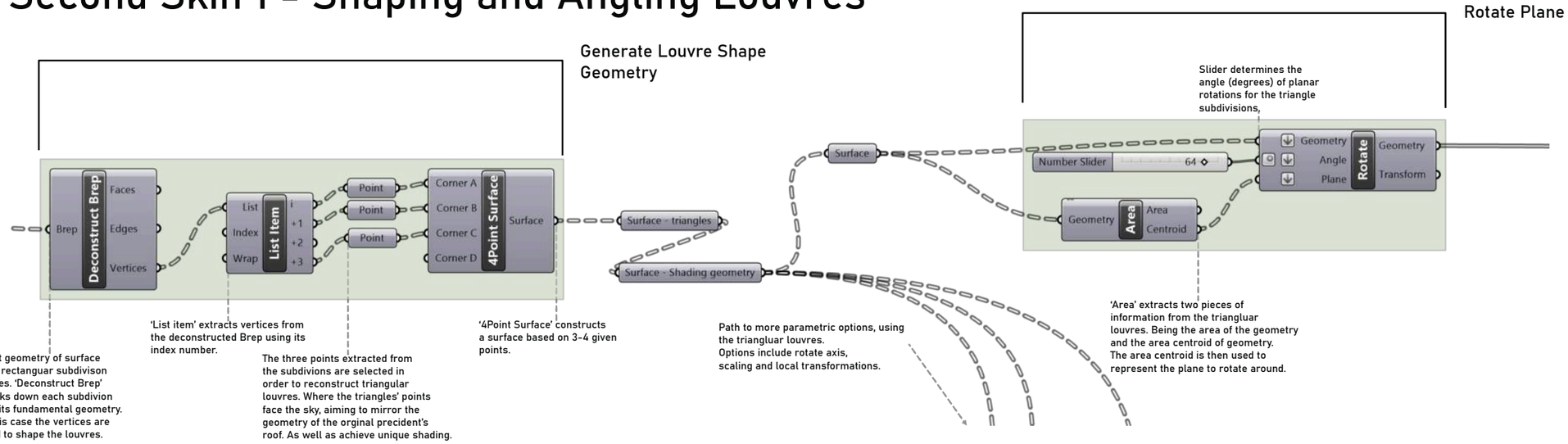
Grasshopper Script

Massing Geometry - Angling Roof Volumes



This section of the grasshopper massing script is responsible for the angling of the roof geometry and is repeated for all volumes around the main box geometry of my precident. This portion highlights how vital parametric design tools such as grasshopper are in rapidly and effectively testing key design concepts.

Second Skin 1 - Shaping and Angling Louvres



This portion of the grasshopper second skin script controls louvre shape and rotation. A key element of the set up in the script that promotes effective testing of preformance based design factors is the definition of progressive geometric changes. The "Surface - Shading geometry" node allows for many variations to stem from the shaping step.

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